# CrimeVision AI - Intelligent Multimodal Forensic Analysis Platform

Skill Wallet Capstone Project 2 | Category: Generative AI, Criminal Justice & Public Safety

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## Overview

CrimeVision AI is a Streamlit app that combines image, audio and video evidence into one forensic timeline. It also provides an evidence chat that answers only from the collected evidence, and exports a PDF report.

## Features

- **Scene Evidence Tagger:** simulated RT-DETR detection of persons, firearms, edged weapons, blood spatter and forced entry points, with threat level, scene classification and investigative leads.
- **Witness Audio:** Whisper transcription (via Groq API) with key events and timestamps.
- **CCTV Video:** frame sampling and movement tracking with event classification (OpenCV).
- **Fusion and Evidence Chat:** a master ledger with evidence IDs, and Q&A that cites those IDs and replies 'Insufficient evidence in ledger' when the answer isn't there.
- **PDF Report:** ReportLab export with chain-of-custody details.
- **Demo mode:** works with no API key.

## Tech Stack

Python, Streamlit, OpenCV, Pillow, ReportLab, Whisper (Groq), Groq / Gemini LLM.

## Run Locally

```
pip install -r requirements.txt
streamlit run app.py
```

Optional: enter a Groq or Gemini API key in the sidebar for live AI mode. Never commit keys to the repo.

## Live Demo

Add your Streamlit Cloud link here.

## Screenshots

Add screenshots of each tab here.

## Disclaimer

Automated analysis is investigative support only and must be verified by a qualified examiner.